

## SELMA 6 SSE, AL, USA

WMO: 750100

Lat: 32.3355N

Lon: 86.9787W

Elev: 157

StdP: 14.61

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 63897

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 22.5       | 26.1 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 21.1                  | 46.5 | 19.3 | 46.8 | 4.3                   | 130  | 0.362 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 19.2                   | 95.8            | N/A  | 93.7 | N/A  | 91.8 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 6.9                  | 280  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |     |            |     |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-----|------------|-----|-------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |     | n=50 years |     |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max | Min        | Max |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n) | (o)        | (p) |       |  |
| 17.5              | 15.0 | 12.9 | DB                         | 16.3 | 97.6               | 4.2 | 2.5                                                | 13.2 | 99.4       | 10.7 | 100.9      | 8.3 | 102.3      | 5.3 | 104.1 |  |
|                   |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A | N/A        | N/A | N/A   |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 64.7   | 45.6  | 51.1  | 57.0  | 63.9  | 71.9  | 79.2  | 80.6  | 80.5  | 76.1  | 65.3  | 53.8  | 50.1  |
|                                                                   | DBStd       | 14.40  | 10.52 | 10.03 | 8.72  | 6.77  | 5.65  | 3.45  | 2.88  | 3.56  | 4.84  | 8.01  | 8.38  | 9.74  |
|                                                                   | HDD50       | 538    | 215   | 102   | 36    | 3     | 0     | 0     | 0     | 0     | 0     | 3     | 52    | 127   |
|                                                                   | HDD65       | 2308   | 605   | 400   | 271   | 100   | 17    | 0     | 0     | 0     | 1     | 97    | 347   | 470   |
|                                                                   | CDD50       | 5888   | 78    | 133   | 253   | 419   | 678   | 876   | 948   | 946   | 784   | 476   | 167   | 130   |
|                                                                   | CDD65       | 2183   | 3     | 11    | 22    | 67    | 230   | 426   | 483   | 481   | 335   | 105   | 12    | 8     |
|                                                                   | CDH74       | 20433  | 9     | 46    | 179   | 668   | 2188  | 4181  | 4623  | 4561  | 2910  | 969   | 75    | 24    |
|                                                                   | CDH80       | 8840   | 0     | 1     | 18    | 125   | 823   | 2020  | 2138  | 2191  | 1243  | 273   | 8     | 0     |
|                                                                   | WSAvg       | 5.1    | 6.2   | 6.4   | 6.5   | 5.9   | 4.9   | 4.5   | 4.3   | 4.1   | 3.9   | 4.7   | 4.8   | 5.6   |
| Precipitation                                                     | PrecAvg     | 52.0   | 5.3   | 5.1   | 6.1   | 4.3   | 3.5   | 4.2   | 4.6   | 4.1   | 3.7   | 2.7   | 3.9   | 5.0   |
|                                                                   | PrecMax     | 73.0   | 12.2  | 14.0  | 13.0  | 13.6  | 9.5   | 10.4  | 11.5  | 9.5   | 10.9  | 9.3   | 10.4  | 20.4  |
|                                                                   | PrecMin     | 34.2   | 1.1   | 1.5   | 1.1   | 0.2   | 0.1   | 0.1   | 0.5   | 0.7   | 0.4   | 0.1   | 0.6   | 1.3   |
|                                                                   | PrecStd     | 9.1    | 2.4   | 2.5   | 3.3   | 2.7   | 2.4   | 2.6   | 2.2   | 2.0   | 2.5   | 2.0   | 2.4   | 3.1   |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 75.3  | 78.8  | 82.4  | 85.9  | 92.5  | 97.9  | 96.5  | 100.8 | 95.4  | 89.9  | 80.4  | 77.6  |
|                                                                   |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 2%          | DB     | 71.0  | 75.1  | 78.7  | 83.6  | 89.8  | 95.0  | 94.2  | 96.1  | 92.5  | 86.0  | 75.8  | 73.4  |
|                                                                   |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 5%          | DB     | 66.9  | 71.1  | 75.5  | 81.2  | 87.3  | 92.8  | 92.4  | 93.3  | 90.2  | 83.1  | 72.9  | 70.0  |
|                                                                   |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 10%         | DB     | 62.9  | 67.7  | 72.1  | 78.2  | 85.1  | 90.6  | 90.3  | 90.6  | 87.4  | 79.8  | 69.5  | 66.6  |
|                                                                   |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 21.6  | 22.5  | 23.3  | 24.3  | 23.3  | 21.1  | 19.2  | 19.6  | 21.0  | 24.2  | 25.4  | 20.1  |
|                                                                   |             | MCDBR  | 23.0  | 23.1  | 24.7  | 26.2  | 26.6  | 24.5  | 21.9  | 23.8  | 24.2  | 26.4  | 26.5  | 21.7  |
|                                                                   |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                   |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.324 | 0.334 | 0.361 | 0.395 | 0.445 | 0.478 | 0.541 | 0.530 | 0.441 | 0.375 | 0.347 | 0.326 |
|                                                                   | taud        |        | 2.525 | 2.501 | 2.428 | 2.325 | 2.232 | 2.184 | 2.019 | 2.046 | 2.280 | 2.447 | 2.466 | 2.533 |
|                                                                   | Ebn at Noon |        | 285   | 293   | 291   | 284   | 269   | 259   | 242   | 243   | 262   | 275   | 273   | 278   |
|                                                                   | Edh at Noon |        | 27    | 30    | 35    | 41    | 45    | 47    | 55    | 53    | 40    | 32    | 28    | 25    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 881   | 1062  | 1428  | 1786  | 1968  | 1972  | 1903  | 1767  | 1572  | 1330  | 999   | 757   |
|                                                                   | RadStd      |        | 64    | 133   | 131   | 133   | 135   | 151   | 128   | 104   | 127   | 161   | 96    | 80    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (40 neighbors) | +0.83 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +259  | +167  |

Nomenclature: See separate page